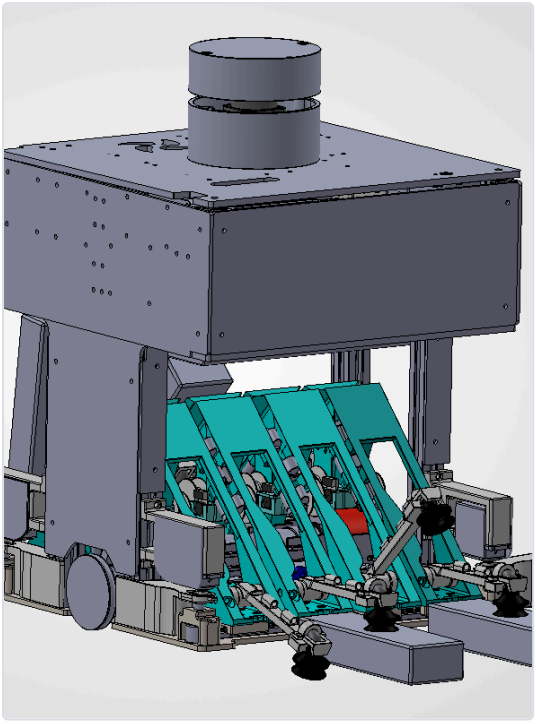
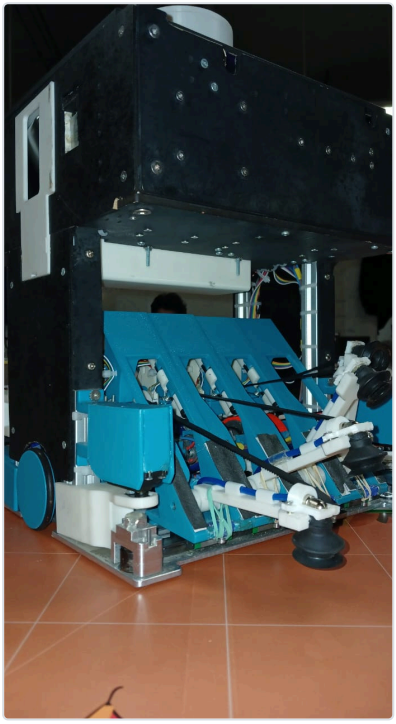
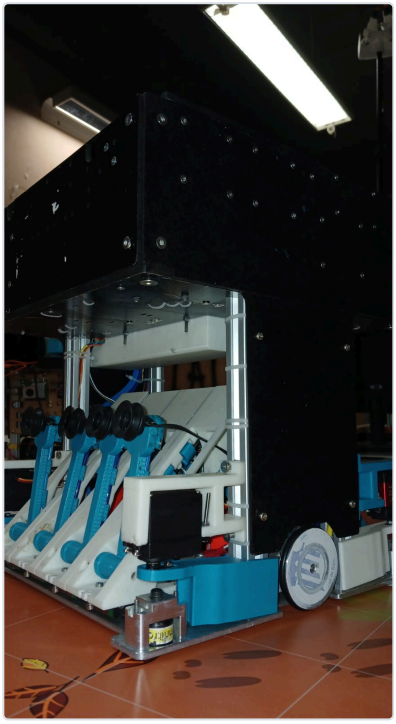
AEROBOTIX INSAT · EUROBOT 2026 · CAD TEARDOWN

How the robot in “Slide final CAD” is built

A walkthrough of the drivetrain, the eight suction picking modules, the sliding (rail) mechanism that spaces them, the vacuum system, the frame and the electronics. Every finding comes from reading the STEP file part by part, with labelled renders taken straight from the CAD.







~300 × 300
footprint (mm), about 435 mm tall with the LiDAR

16
motors and servos: 2 drive, 8 arm, 6 servos (plus 2 pumps)

8
suction picking arms, 4 on each face

196 / 415
unique parts / placed instances in the STEP

1. How this was analysed
2. Overall layout
3. Motor list
4. Drivetrain and odometry
5. Picking modules
6. The sliding (“glider”) mechanism
7. Vacuum / pneumatics
8. Frame, panels and how they are fixed
9. Electronics
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1. How this was analysed

The file is a SolidWorks 2026 STEP AP214 export (88 MB). It holds 196 unique parts placed 415 times. A script read the assembly tree to find each part's exact position and size, then drew wireframe views from the edges of the real geometry. Every picture in this document comes from that process.

> **Limits.** The STEP holds **no material data**, and every part has the same default colour. Materials below come from the part names and the two photos. Statements the CAD can't confirm directly are marked INFERRED.

Reading the pictures

Drive / odometry · Picking modules, rails · Suction / pneumatics · Servos and linkages · Electronics · Ball casters · Crates (game element) · Structure

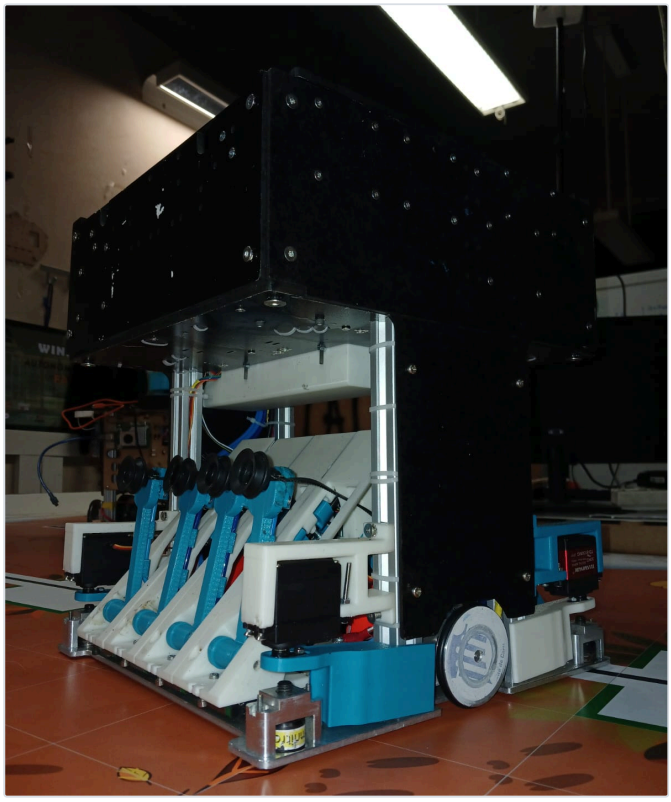
In views that study one subsystem, that subsystem is drawn red and everything else light grey. A 50 mm scale bar sits at the bottom of each view.

Face A / Face B. The robot has two identical picking faces, front and back. In the photos one face is teal and the other white. The CAD alone can't tell which is which, so this document calls them Face A and Face B.

The real robot



Teal face. You can see the 4 inclined chutes, the black timing belts driving the arms, and the single black side panel on the left.



White face. The 4 arms are raised with their suction cups up. You can see the aluminium base plate, the caster blocks and the white LiPo drawer under the top box.

2. Overall layout

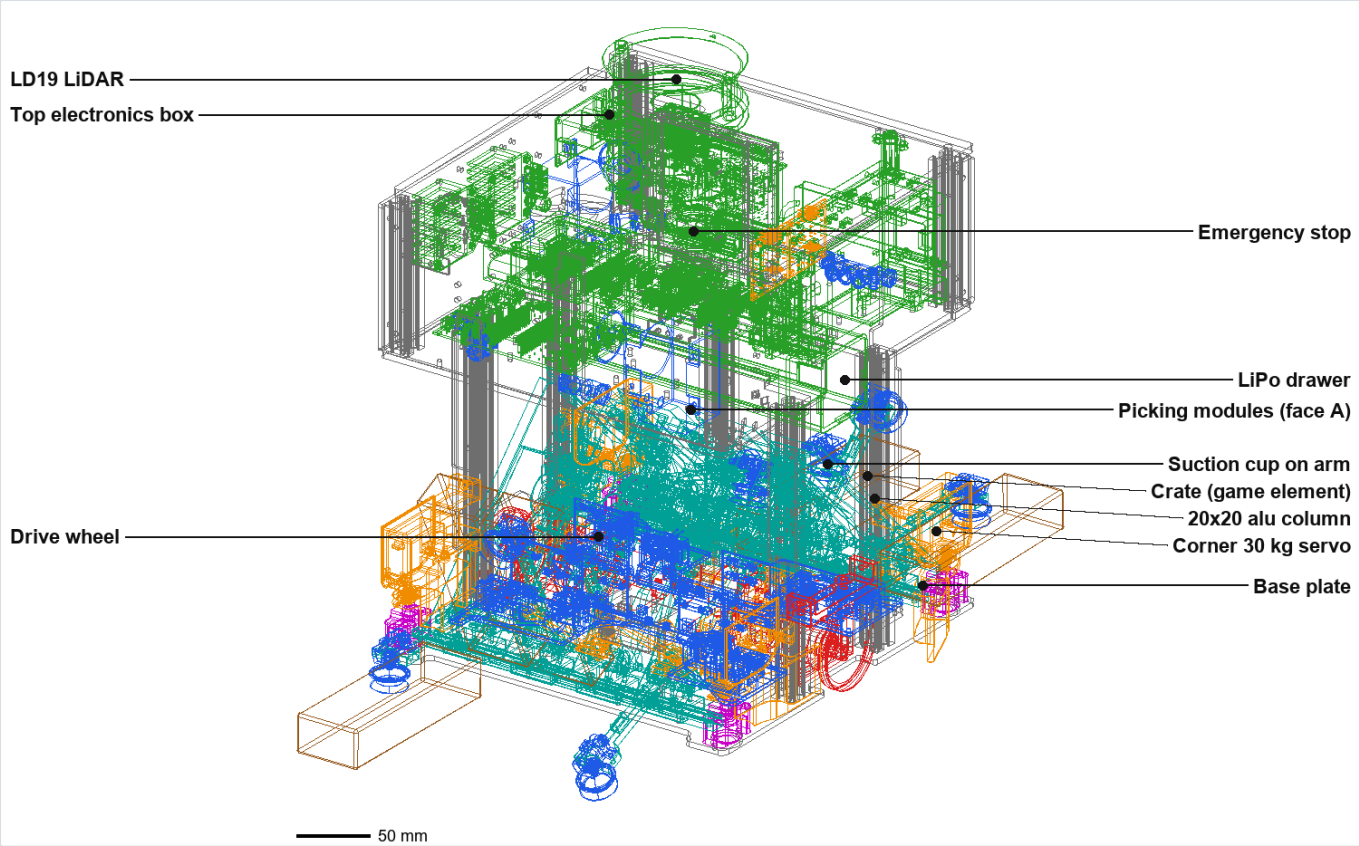


Figure 2.1: the whole robot as wireframe, colour-coded by subsystem.

Level	Height	What lives there
Base	0 to 8 mm	Aluminium base plate, two weight plates, the two drive gearmotors, the two MGN12 rails, eight solenoid valves
Middle	8 to 219 mm	Eight picking modules (4 per face), the arms, the servos that space the modules, four 20×20 columns
Top box	219 to 342 mm	All electronics, the 4S LiPo (in a drawer), the power bank and both vacuum pumps
Mast	342 to 426 mm	Two-part printed tower holding the LD19 LiDAR; the emergency stop sits on the lid

The robot is **symmetric front to back**, so it can pick with either face without turning around. The drive axle runs through the exact centre of the robot, so it spins on the spot.

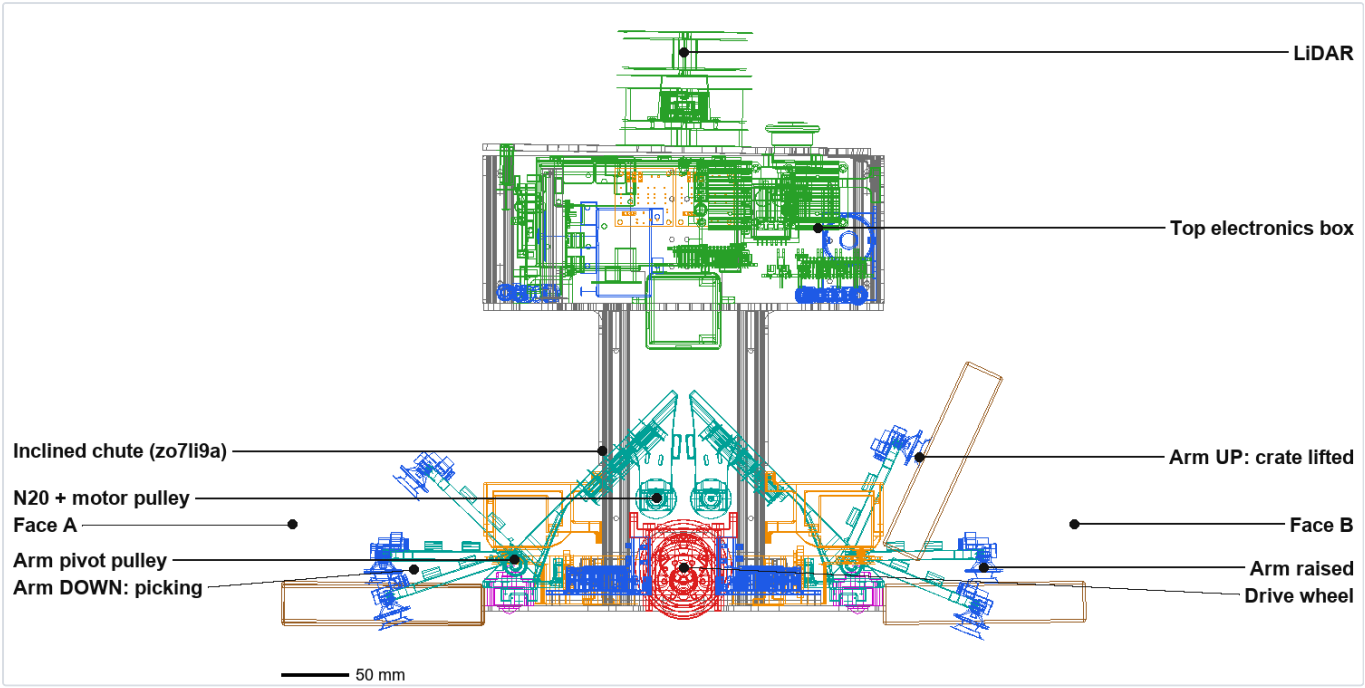


Figure 2.2: side view (looking across the robot). The arms are saved in three poses: flat on a crate (picking), raised, and lifting a crate clear of the floor. The chutes lean in at about 45°, and the top box sits on two narrow columns.

3. Motor list

Actuator	Qty	Where	Job
Pololu 37D-size metal gearmotor, 12 V, in a JGB37-520 bracket	2	Centre of the base, back to back	Differential drive, direct to the wheels
N20 micro metal gearmotor	8	High inside each chute	Turns one picking arm through a belt
TIANKONGRC 30 kg servo	4	At both ends of each rail, shaft vertical	Swings the "organiser" lever that pushes the modules back together
S0900M 9 kg servo	2	Centre of each face, behind the rail	Turns the "separation arms" that push the modules apart
CK501-1BPM 12 V vacuum pump	2	Top box	Vacuum for one face's 4 cups
Fa0520F 12 V solenoid valve	8	Base level, near the arms	Grab/release for one cup

Motor drivers

- **Drive motors:** 2× IBT-2 (BTS7960) H-bridges, one per motor.
- **N20 arm motors:** 2 custom "ESP motors 2.0" boards. Each has an ESP32 and two TB6612FNG dual drivers, which gives 4 motors per board and 8 in total.
- **Servos:** 2 custom servo boards.
- **Valves and pumps:** a custom "Carte Mosfet 2.0" switching board.

4. Drivetrain and odometry

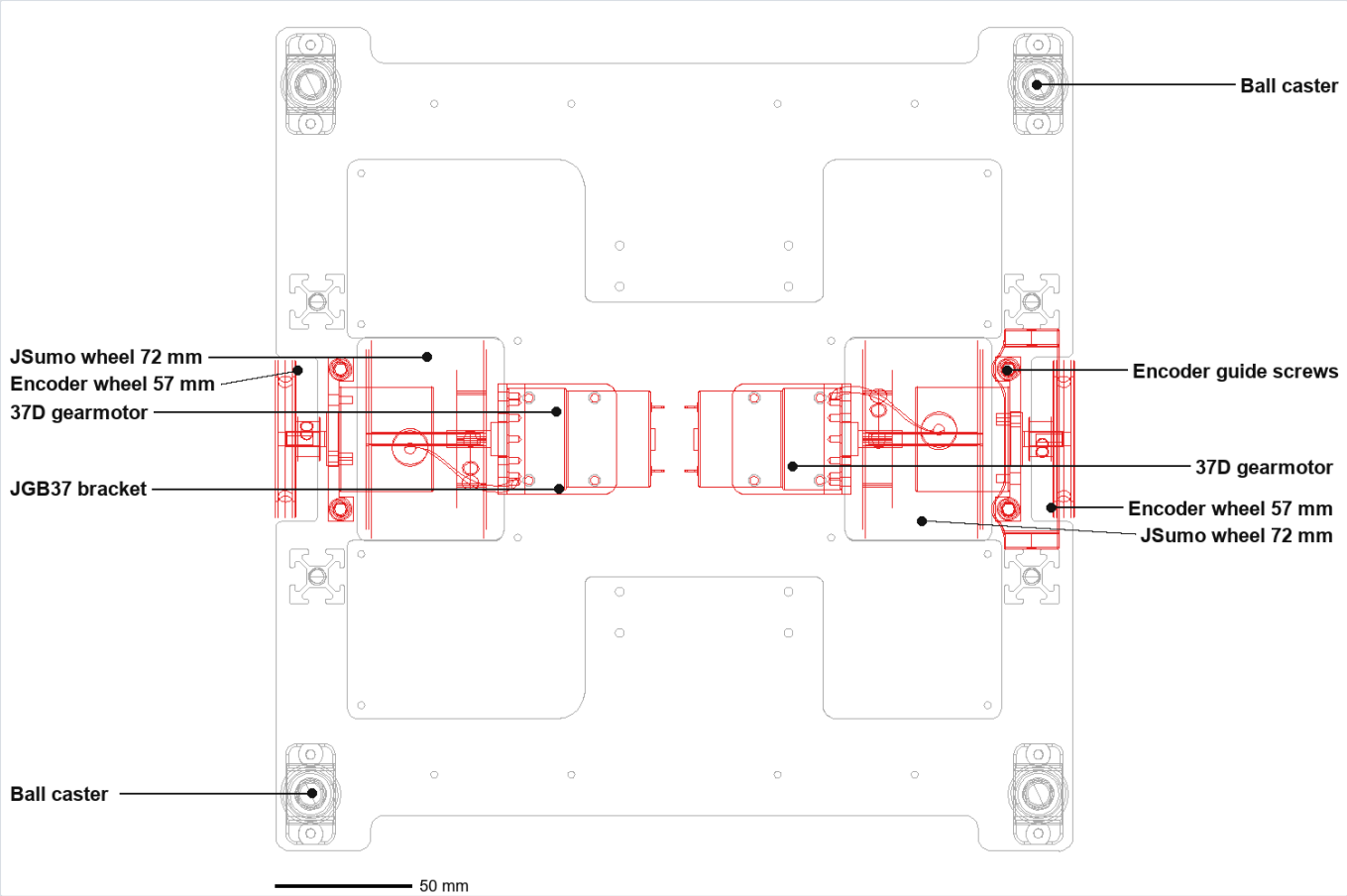


Figure 4.1: drivetrain seen from below (drive parts in red). The two gearmotors sit back to back across the middle, each driving its wheel directly. A small encoder wheel sits just outside each drive wheel. There is a ball caster in each corner.

- **Motors:** two 37 mm gearmotors (about 79 mm long) held by JGB37-520 L-brackets, lying across the robot and back to back. Each **drives its wheel directly**, with no belt or gearbox in between.
- **Wheels:** JSumo, Ø72 mm, about 41 mm wide, about 180 mm between wheel centres. The axle crosses the centre of the robot, so it turns on the spot.
- **Casters:** 4 Omnitrack ball casters (Ø20 ball) in milled corner blocks.
- **Odometry:** separate **Ø57 mm encoder wheels**, one outside each drive wheel and on the same axis. They roll freely, so they don't slip when the drive wheels spin or skid.

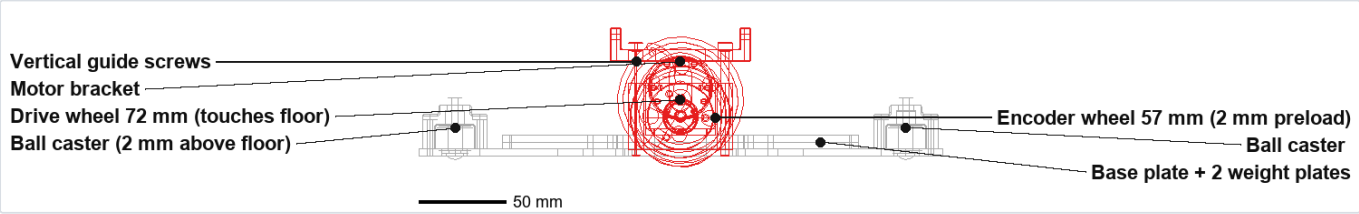


Figure 4.2: drivetrain side view, showing the height of each floor contact.

> **A detail worth copying: the heights.** The drive tyres reach 8 mm below the base plate, which is the floor line. The ball casters stop **2 mm above** the floor, so the drive wheels always carry the weight and keep grip. The encoder wheels go **2 mm below** the floor in the CAD. Each sits on a block that slides up and down on two 65 mm guide screws, so on the real floor it is pushed up and pressed down by spring force INFERRED, keeping constant contact for accurate position tracking.

5. Picking modules (“zo7li9a”)

Each face has **four identical modules** side by side, so 8 in total. A module is a 3D-printed triangular chute about 51 to 55 mm wide that leans about 45° toward the centre of the robot. It carries three things:

- its own **N20 gearmotor**, mounted high on the inside of the chute, with a Ø30 mm pulley;
- a **picking arm** about 107 mm long that pivots at the bottom outer corner of the chute. A pulley is built into the pivot, and the tip carries a bellows suction cup (about Ø28) on a 90° fitting;
- an adapter plate bolted onto an **MGN12C carriage**, so the whole module can slide sideways (see section 6).

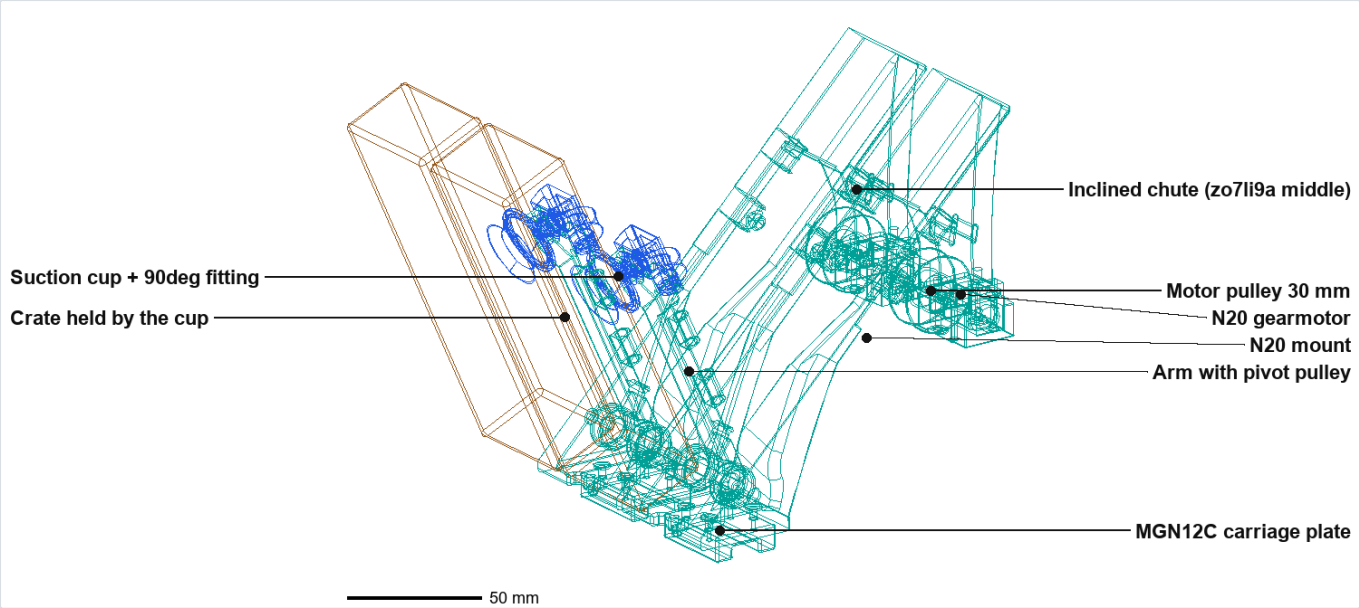


Figure 5.1: two neighbouring modules from Face B. The arm has lifted a crate (50 × 30 × 150 mm, named “hazel nut crate” in the CAD) off the floor with its suction cup.

Drive path: N20 motor → 30 mm pulley → timing belt down along the slope → pulley on the arm pivot. The belt isn't modelled in the CAD, but it shows clearly in the teal photo as black belts running diagonally.

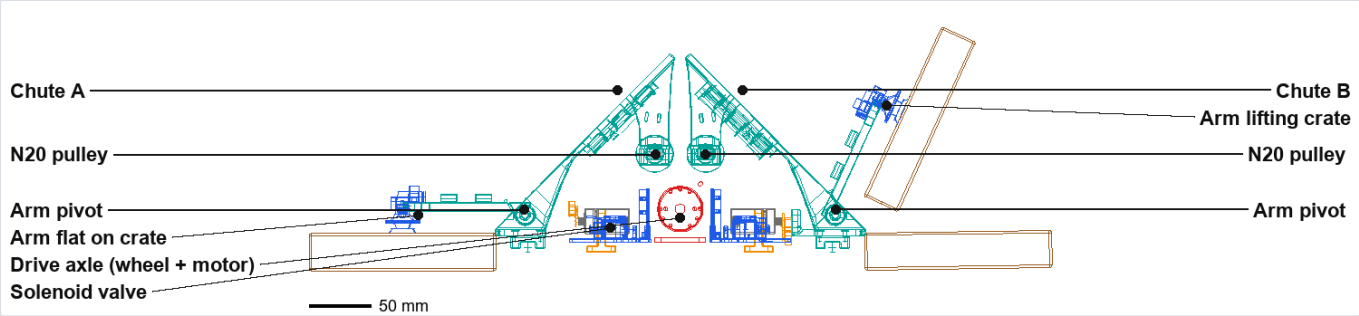


Figure 5.2: cut through the middle of the robot. Face A (left) has its arm flat on a crate. Face B (right) has its arm swung up holding a crate at about 60°. Both chutes, their motor pulleys and the arm pivots are visible. The drive axle sits in the middle, and the solenoid valves sit low beside it.

Picking sequence (from the saved poses)

1. The arm lies flat in front of the robot and the cup lands on top of the crate.
2. The valve closes, the cup grips, and the N20 turns the arm upward around its pivot.
3. The crate swings up and is held clear of the floor, leaning against the chute. All four modules on a face work independently, so a face can collect up to four crates.

The arm part is called “picking arm pulley double action”. That suggests it is meant to turn both ways (down to pick, and up past the chute to carry or flip a crate) INFERRED.

6. The sliding (“glider”) mechanism

This is the cleverest part of the design. The four modules on each face are **not fixed**. Each rides on its own carriage on a shared linear rail, so the robot can change the gap between them, for example to match how the crates are laid out on the table, or to pack tight. INFERRED purpose.

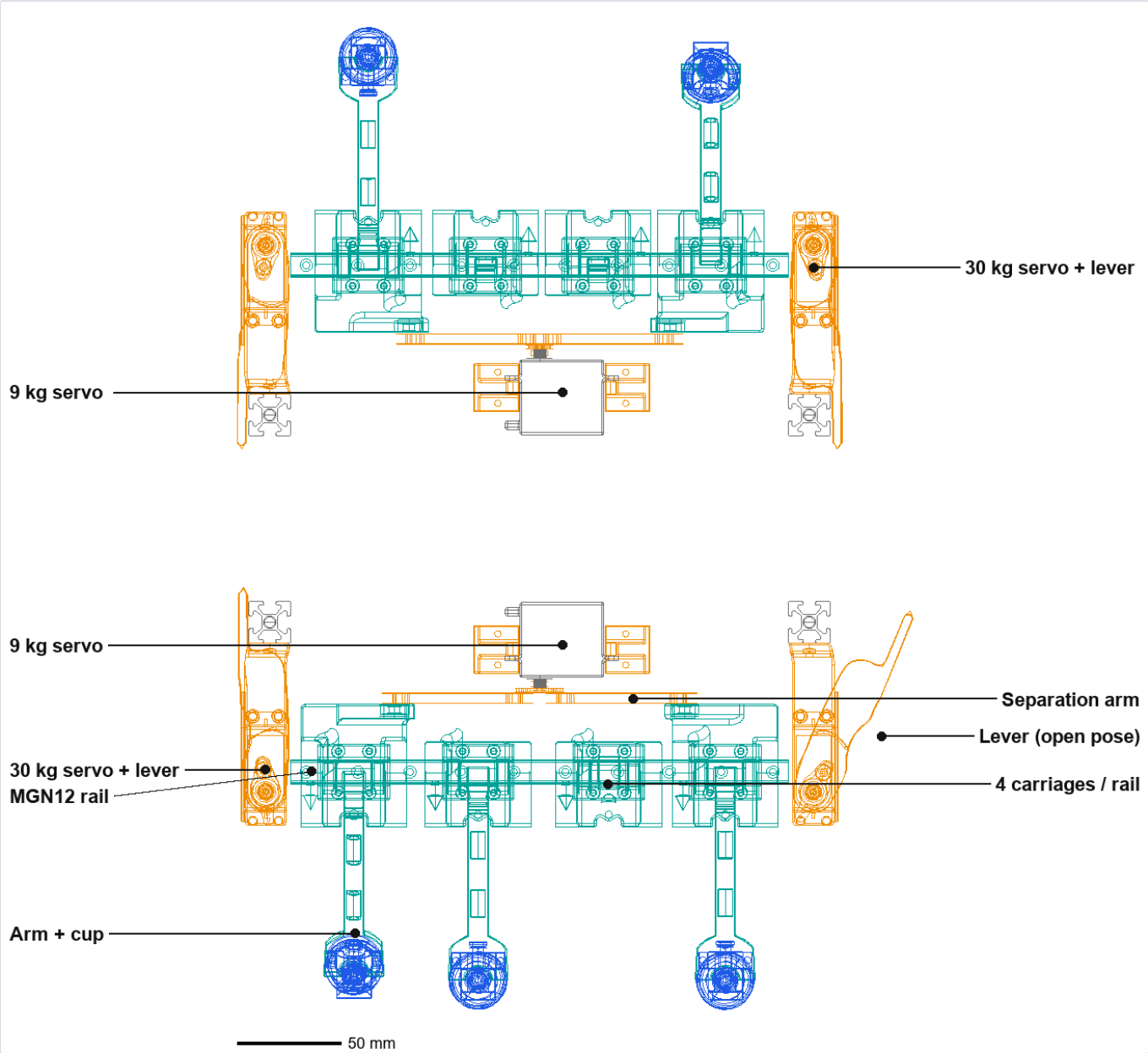


Figure 6.1: top view cut at floor level. Each face has one MGN12 rail carrying 4 module carriages. A 30 kg servo with a long lever sits at each end of the rail, and a 9 kg servo sits in the middle. One lever is saved in its swung-open pose.

The parts

1. **Rail.** One **MGN12 linear rail**, 240 mm long, is screwed straight onto the base plate along each face (12 × 8 mm section). Each module is bolted to its own **MGN12C carriage**, so it slides sideways almost without friction.
2. **Spreading (9 kg S0900M servo).** It sits in the middle behind the rail. It turns two 72 mm “separation arms” that push the **outer** modules outward. Each outer module carries a **small bearing (5 mm bore, 13.8 mm outside) as a cam follower**, so the arm rolls on it instead of rubbing.
3. **Compacting (30 kg servo at each end).** The shaft points vertically, and the servo swings a long “organiser” lever flat across the floor to push the stack back toward the middle.

4. **Equal spacing (folding chains).** Neighbouring modules are linked by small chains (next page). They make the inner modules follow the outer ones and stop every gap at the same maximum.



Figure 6.2: 3D view of the spacing mechanism (red) on the base plate and rails (grey). The bearings the separation arms push on are visible, and so is the lever saved in its open pose.

Measured from the CAD	Value
Module width	51 to 55 mm
Distance between module centres, packed tight	about 54 mm
Distance between module centres, in the spread pose saved in the CAD	about 56 to 63 mm
Largest spacing the 240 mm rail allows	about 69 mm INFERRED

Folding chains between modules

In each of the three gaps between neighbouring modules there is a **small chain**: three links (“zo7li9a chain tooth”) joined by pins (“smooth stem tooth”, 26 mm and 69 mm long). The pins run along the 45° slope, so the chain folds **sideways**. When the modules move apart, the chain straightens and stops at its full length. When the modules are pushed together, it folds up into the gap.

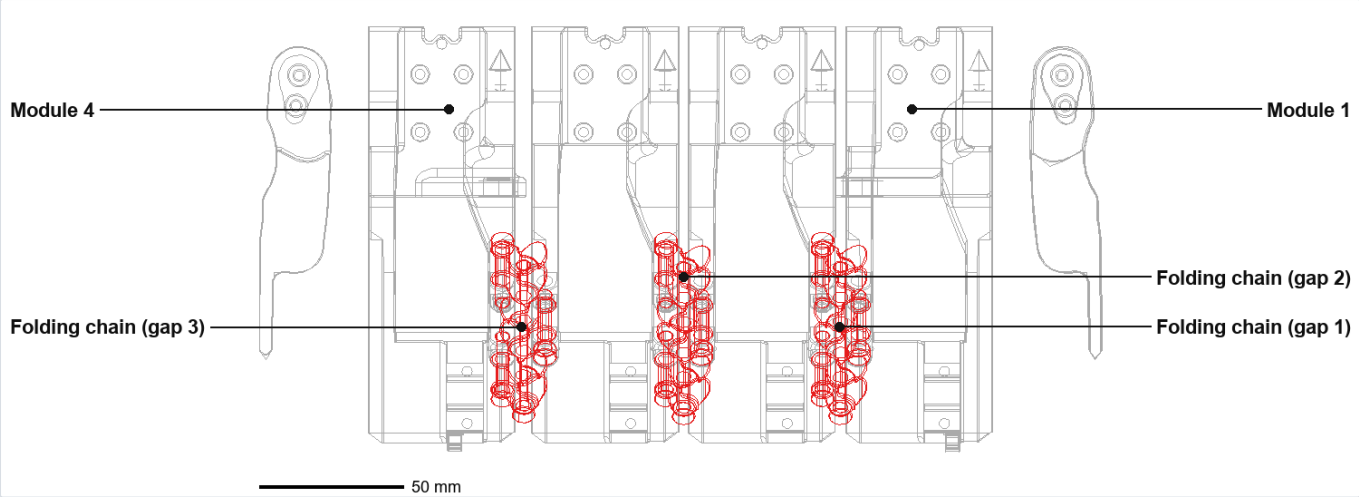


Figure 6.3: top view of the four Face-B chutes with the chains (red) sitting in the three gaps.

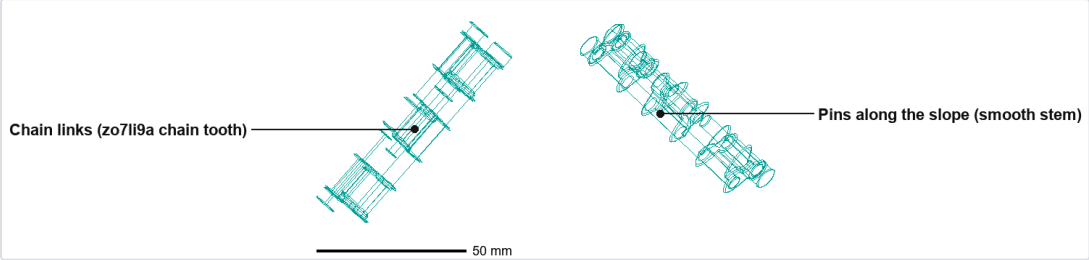


Figure 6.4: one gap on its own: the pins (left) and the links (right).

> The geometry clearly shows a folding chain linking neighbouring modules. That it acts as an **end stop that sets the largest gap**, so all four modules end up evenly spaced, is the most likely reading, but it is INFERRED. It's worth asking the team who designed it.

7. Vacuum / pneumatics

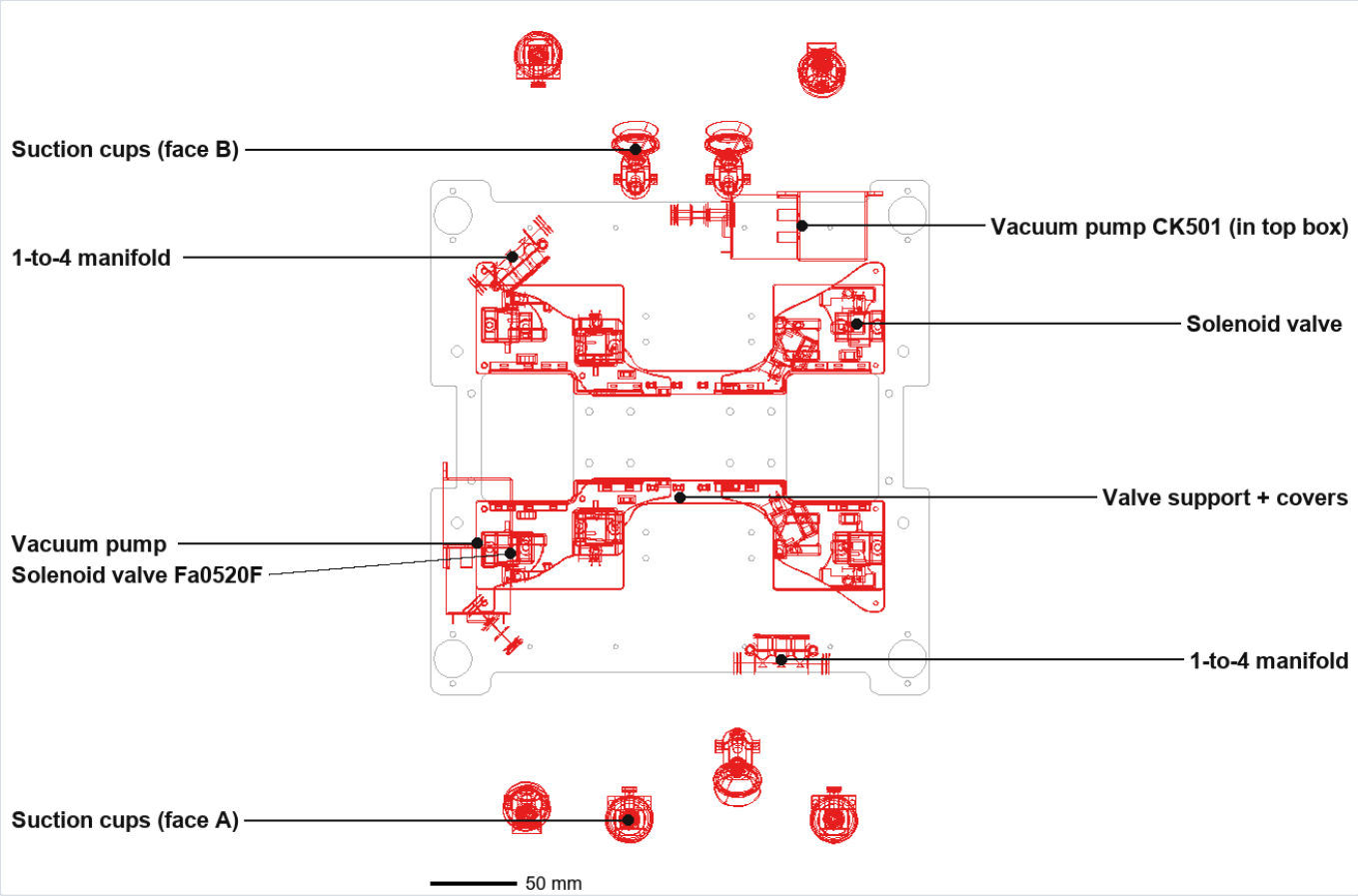


Figure 7.1: pneumatic parts (red), seen from above. The pumps sit in the top box. Valves, splitters and cups are at the bottom.

- **Pumps:** 2× **CK501-1BPM 12 V** vacuum pumps, one per face, inside the top box.
- **Splitters:** each pump feeds a **1-into-4 manifold** (4 mm tubes). There is also a 6-to-8 mm adapter on each line.
- **Valves:** 8× **Fa0520F 12 V solenoid valves**, one per cup, mounted low on printed supports near the arms under printed covers (`ev1` to `ev8`). Each cup can grab and let go on its own.
- **Control:** the custom MOSFET board switches the valves and pumps.

8. Frame, panels and how they are fixed

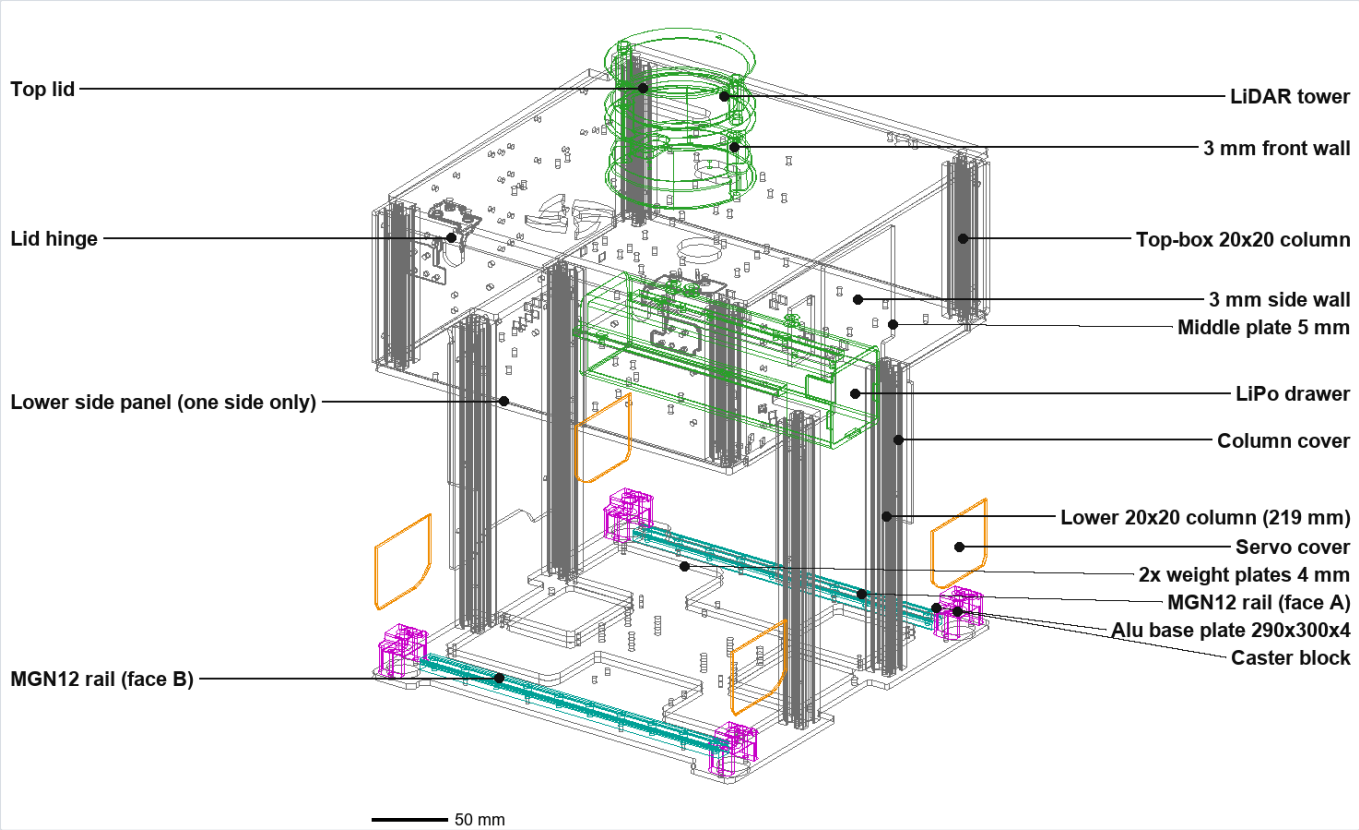


Figure 8.1: the structural parts only, in 3D.

Part	Size (from CAD)	Material and how it is fixed
Base plate	290 × 300 × 4 mm	Aluminium (silver in the photos). **The backbone: everything mechanical bolts to it.** That includes the rails, motor brackets, caster blocks, valve supports, servo holders and the four columns. It has cut-outs for the wheels.
Weight plates	2 × 238 × 204 × 4 mm	Stacked on the base under the motors. Named "weighted base", so most likely steel, there to bring the centre of gravity down on a top-heavy robot INFERRED.
Caster blocks	18 × 25 × 37 mm	Milled aluminium blocks at the 4 corners (visible in the white-face photo).
MGN12 rails	240 mm, 2×	Screwed straight onto the base plate along the front and back edges.
Lower columns	20×20 extrusion, 219 mm, 4×	Aluminium profiles standing on the base near the middle of each side. They carry the middle plate. Thin 2.7 mm cover plates hide them.
Middle plate	286 × 296 × 5 mm	The floor of the top box. The LiPo drawer passes through a cut-out in it.
Top-box columns	20×20 extrusion, 110 mm, 4×	One in each corner of the box.
Box walls	3 mm sheets	Black in the photos, likely acrylic or PVC sheet INFERRED. They sit against the corner profiles and are screwed into them.
Top lid	288 × 288 × 8.6 mm	**Hinged on the back wall** (2 hinges plus a lift tab), so the lid opens for access to the electronics. It carries the LiDAR mast and the e-stop.
Lower side panel	3 mm, one side only	The black "leg" in the photos. It shows on the left from one face and on the right from the other.
Printed parts	n/a	Chutes, arms, motor mounts, servo holders and covers, valve covers, LiPo drawer, LiDAR tower, and 39 zip-tie anchors. Teal on one face, white on the other (likely PLA/PETG).

> **Order of assembly:** base plate (weight plates, rails, drivetrain, valves and modules all mounted on it) → four columns → middle plate → box profiles and walls → hinged lid with LiDAR. Because the lid is hinged, the electronics can be reached without taking the robot apart.

9. Electronics (top box)

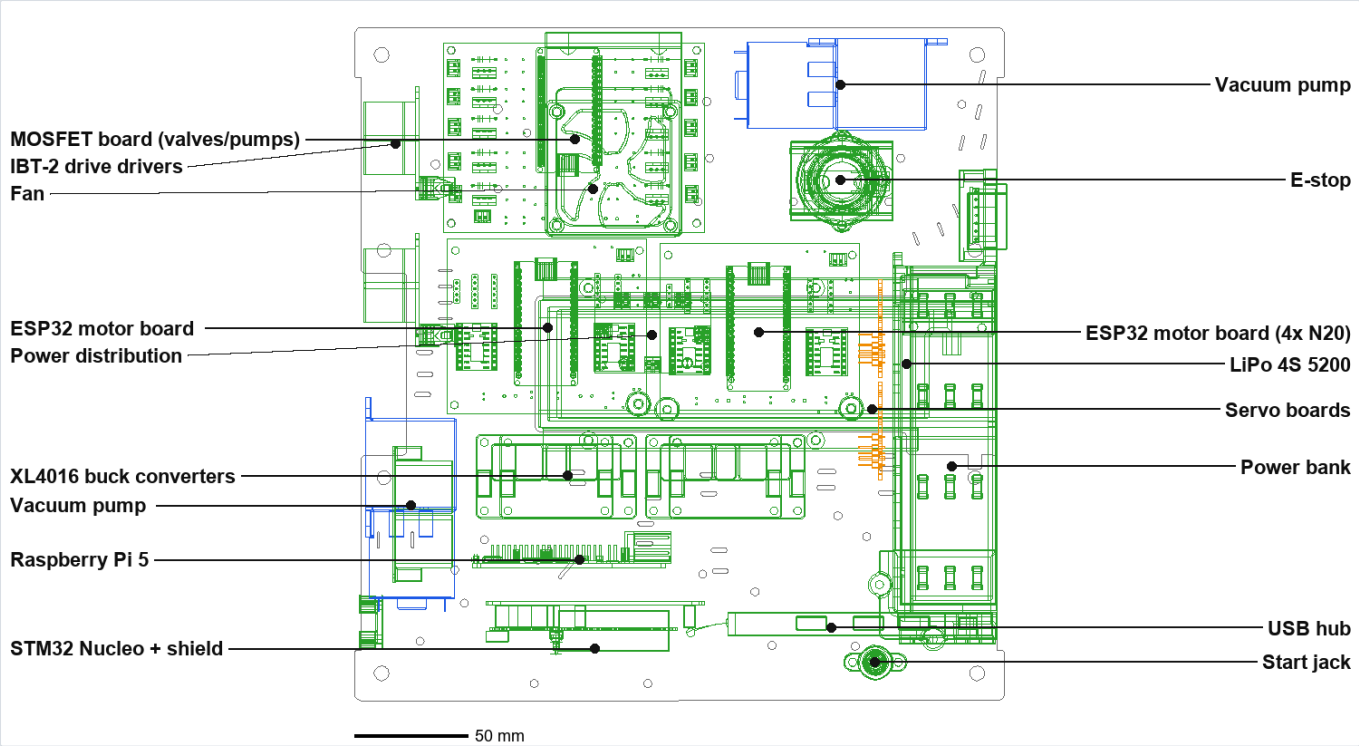


Figure 9.1: top view of the box contents (boards are stacked, so some overlap).

Group	Parts
High-level control	Raspberry Pi 5 with an LD19 LiDAR on the mast
Low-level control	STM32 Nucleo (F401/F411) with a custom shield; a third ESP32 module elsewhere in the box
Motor driving	2× IBT-2 (drive); 2× “ESP motors 2.0” boards (4 TB6612FNG in total, 8 N20s)
Actuator I/O	2× servo boards; “Carte Mosfet 2.0” for the valves and pumps
Power	4S 5200 mAh LiPo in a pull-out drawer, with a voltage checker on the side; a separate power bank (probably to keep the Pi 5's supply clean INFERRED); 5× XL4016 step-down converters; a power distribution board
Other	USB hub, cooling fan, emergency stop on the lid, start-cord jack

10. Takeaways for our robot

1. **Symmetric two-face design.** Identical picking faces on both ends, so the robot never has to turn around to pick.
2. **Separate encoder wheels on vertical guides.** The odometry wheels are independent of the drive wheels and pressed onto the floor. The casters are set 2 mm high so the drive wheels never lose traction.
3. **Each module has its own motor and valve.** 8 small N20 belt-driven arms are simpler and more independent than one big shared mechanism.
4. **Modules on linear rails with adjustable spacing.** MGN12 rails, a servo to spread, servos to compact, and folding chains to keep the gaps even. Only 6 servos control 8 modules' positions.
5. **Low weight, high electronics.** Steel-looking weight plates at the bottom balance a heavy electronics box and LiDAR mast. The hinged lid gives quick access.
6. **Easy battery swap.** The LiPo slides out of a side drawer.

> **Open questions to check with the team or the real robot:** what the folding chains really do (section 6); the N20 gear ratio and belt type; what the walls and weight plates are made of; and what the second (“double action”) direction of the arm is for.

Source: `Slide final CAD.STEP` (SolidWorks 2026, exported 2026-10-15), plus the two photos of the real robot. Renders were drawn as wireframes from the STEP edges, and positions were taken from the assembly placements.